Ising Model Applied To Finance

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Bachelor's degree in Physics

April 2026

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1. Introducción

In this practical exercise, the two-dimensional Ising model on a square lattice will be studied through numerical simulations based on Monte Carlo methods. This model, proposed by Lenz in 1920 to study ferromagnetic and paramagnetic systems, lacked an exact solution until Onsager solved it in 1944. Demonstrating the existence of a phase transition at $K_c = \frac{1}{2}\log(1+\sqrt{2}) \approx 0.44$, the Hamiltonian of this system is:

$$H_{Ising,2D} = -J \sum_{\langle i,j \rangle} s_i s_j \tag{1}$$

where J is the positive coupling constant and $\sum_{\langle i,j \rangle} s_i s_j$ is the nearest-neighbor sum of all spins in the lattice. The spins can take values $s_i \in \{+1, -1\}$. It is important to note that periodic boundary conditions will be imposed. Although this model has an analytical solution, the majority of models predicting a phase transition do not have an analytical solution, and to study higher dimensions or other similar problems, it will be necessary to use simulations to obtain results. The purpose of this practical exercise is to provide an introduction to numerical simulations and compare them with theoretical results, as well as to study the algorithms themselves and evaluate the efficiency of each one. Finally, an attempt will be made to explain the behavior of financial markets using this Hamiltonian.

2. Autocorrelation times

In this experiment, we will use the Heat Bath (HB) algorithms, which consist of a Markov chain that changes the configuration of each spin at every step, and the Swendsen-Wang (SW) algorithm, which operates by changing all spin clusters (parallel spin clusters) in each Monte Carlo step to attempt to mitigate the critical slowing down effect in the vicinity of the critical point. We begin by phenomenologically and graphically studying the thermalization (the system is considered thermalized if more than 6 or 7 instantons are observed) of these two algorithms for different lattice sizes and l_{block} values. If the system is thermalized, we can consider that, given an initial configuration, it no longer carries any statistical weight and the simulation correctly represents what occurs in phase space. After a quick check, it is confirmed, as expected, that thermalization occurs much faster for the SW algorithm; with an l_{block} of 100, it thermalizes for small lattices, and with an l_{block} of 1000, it thermalizes for all lattice sizes up to 512. On the other hand, the HB algorithm does not begin to thermalize for small lattices until $l_{block} = 1000$.To study the efficiency of these algorithms more precisely, it is convenient to study the autocorrelation functions; the autocorrelation function describes how the memory of an observation is lost as the simulation progresses [1]. Mathematically, we can express it as

$$\widetilde{C}_{OO}(t) = \langle O_s O_{s+t} \rangle_\beta - \langle O \rangle_\beta^2 \tag{2}$$

Two essential quantities for studying the efficiency of our algorithms are the integrated autocorrelation time and the exponential autocorrelation time. To study these quantities, a simulation is

performed in which we know the system will thermalize. As has been seen, for SW, it will suffice to perform 10000 iterations, whereas for HB, we will perform a minimum of 100000 iterations, and for $L > 32$, we will perform 400000 iterations. Following this, we will be able to obtain the integrated autocorrelation times shown in 1 in order to study the dynamical exponent, which describes how long it takes for a system to return to equilibrium after a fluctuation. This time diverges when we are close to the critical point (critical slowing down); the integrated autocorrelation time is defined as

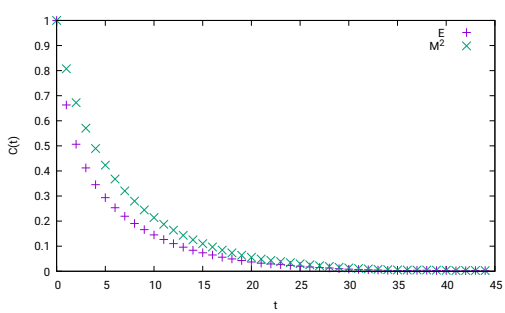
$$\tau_{int,O} = \frac{1}{2} + \sum_{t=1}^{\infty} \frac{C_{OO}(t)}{C_{OO}(0)} \tag{3}$$

where C is the previously described autocorrelation function for a given observable.

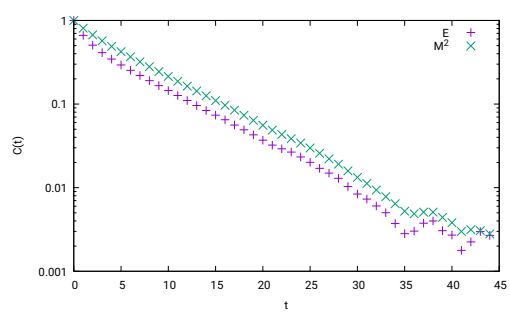
L	τ_E		τ_{M^2}		τ_F	
	HB	SW	HB	SW	HB	SW
8	$4{,}493 \pm 0{,}047$	$2{,}628 \pm 0{,}067$	$6{,}241 \pm 0{,}077$	$2{,}603 \pm 0{,}066$	$6{,}614 \pm 0{,}084$	$1{,}907 \pm 0{,}042$
16	$14{,}54 \pm 0{,}27$	$3{,}197 \pm 0{,}090$	$26{,}52 \pm 0{,}67$	$3{,}114 \pm 0{,}086$	$28{,}68 \pm 0{,}75$	$2{,}017 \pm 0{,}045$
32	$50{,}4 \pm 1{,}8$	$3{,}96 \pm 0{,}12$	$130{,}0 \pm 7{,}3$	$3{,}64 \pm 0{,}11$	$142{,}3 \pm 8{,}3$	$2{,}123 \pm 0{,}049$
64	$178{,}4 \pm 5{,}8$	$4{,}78 \pm 0{,}16$	545 ± 31	$4{,}15 \pm 0{,}13$	567 ± 33	$2{,}218 \pm 0{,}052$
128	517 ± 29	$5{,}74 \pm 0{,}21$	1940 ± 210	$4{,}70 \pm 0{,}16$	2090 ± 230	$2{,}385 \pm 0{,}058$
256	$\sim$	$7{,}09 \pm 0{,}29$	$\sim$	$5{,}78 \pm 0{,}22$	$\sim$	$2{,}637 \pm 0{,}067$
512	$\sim$	$7{,}99 \pm 0{,}35$	$\sim$	$6{,}05 \pm 0{,}23$	$\sim$	$2{,}578 \pm 0{,}065$
1024	$\sim$	$9{,}22 \pm 0{,}44$	$\sim$	$6{,}54 \pm 0{,}26$	$\sim$	$2{,}752 \pm 0{,}072$

Cuadro 1: Integrated autocorrelation time

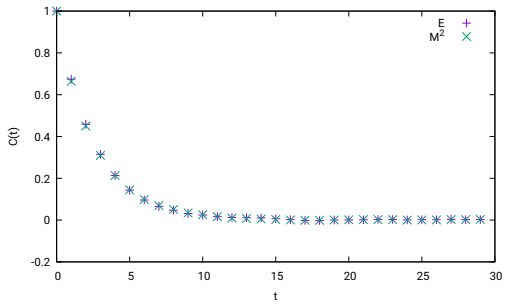
It is clearly observed that for HB, the integrated autocorrelation time increases approximately by a factor of four as we increase the lattice size (since the relationship is $\tau \sim L^z$, and z will be shown to be ~ 2). In contrast, for SW, besides having a smaller autocorrelation time, it does not scale in the same manner but increases much more slowly. To graphically observe the behavior of these autocorrelation times, the autocorrelation functions for the smallest and largest lattices of both models are plotted below.



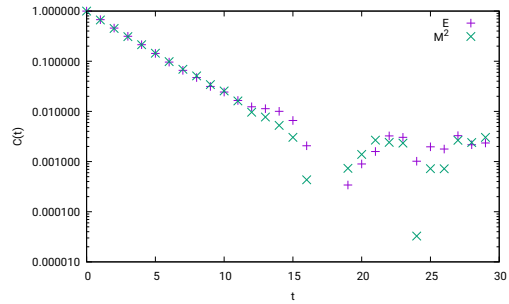
(a) autocorrelation functions L=8 (HB)



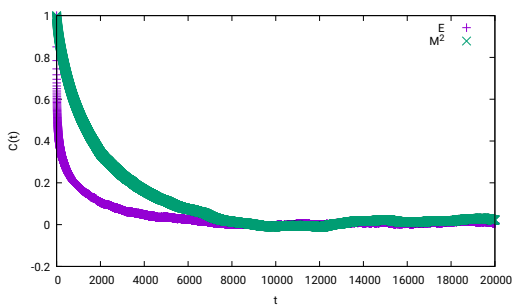
(b) autocorrelation functions L=8 (HB)



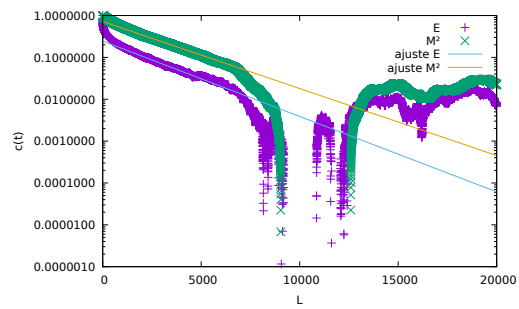
(a) autocorrelation functions L=8 (SW)



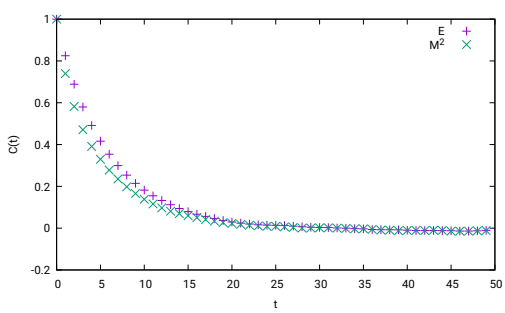
(b) autocorrelation functions L=8 (SW)



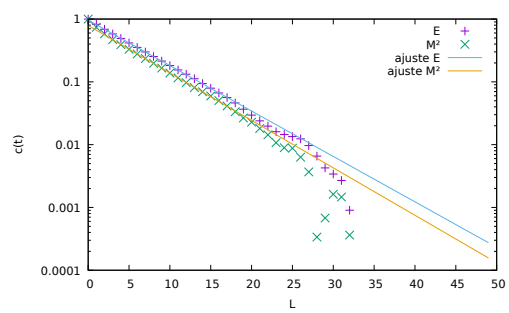
(a) autocorrelation functions L=128 (HB)



(b) autocorrelation functions L=128 (HB)

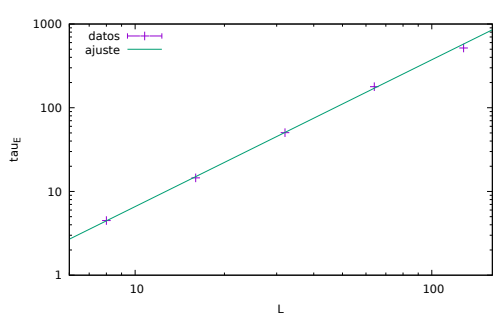


(a) autocorrelation functions L=1024 (SW)

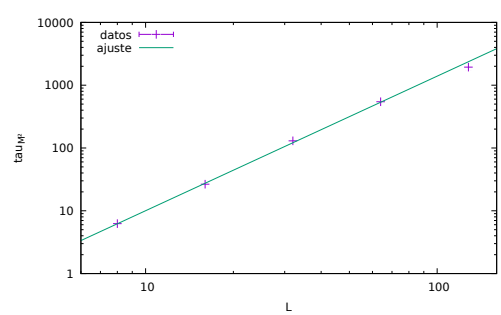


(b) autocorrelation functions L=1024 (SW)

To study the dynamical exponent, the fit ($f(L) = A \cdot L^z$) of the autocorrelation times is plotted for the energy and the squared magnetization in each of the models; it was necessary to remove some parameters in certain fits to obtain a more acceptable P-value.



(a) autocorrelation times for the energy (HB)



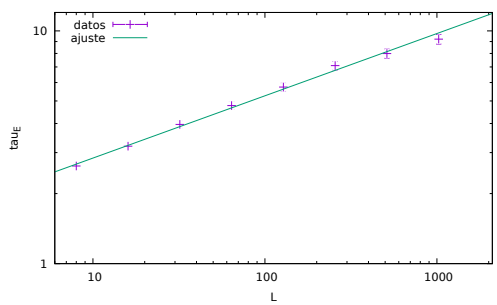
(b) autocorrelation times for the sqaure of the magnetization(HB)

After performing the fit for the energy, I did not obtain an acceptable P-value when using all lattice sizes; however, if I perform the fit without considering the size 16 lattice, I obtain a P-value of 0.064,

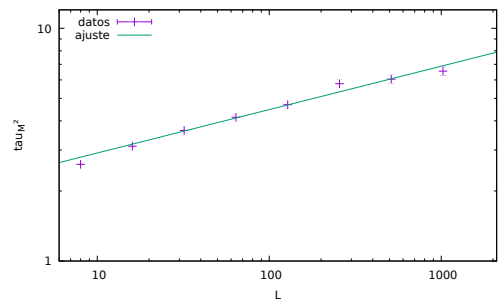
which is almost within the acceptable range. The same occurs for the magnetization, obtaining a P-value of 0.07. The values obtained for the dynamical exponent of the heat bath simulation are

$$z_E = 1{,}744 \pm 0{,}012$$

$$z_M = 2{,}143 \pm 0{,}029$$



(a) autocorrelation times for the energy (SW)



(b) autocorrelation times for the sqaure of the magnetization(SW)

For the Swendsen-Wang algorithm, a similar situation occurs; I obtain consistent results if I disregard the smallest lattices, yielding a P-value of 0.53 and 0.23 for the energy and magnetization fits, respectively.

$$z_E = 0{,}2675 \pm 0{,}0070$$

$$z_M = 0{,}186 \pm 0{,}011$$

To continue with the study of the algorithms, the exponential autocorrelation times are calculated for the largest lattices of each algorithm; mathematically, it is

$$\tau_{exp,O} = \lim_{t\to\infty} \sup \frac{t}{-log\,|C(t)|}, \tag{4}$$

which indicate when the system is thermalized [1], that is, when the simulation can be considered to be showing results from a Boltzmann distribution and has ceased to take the initial configuration into account. The results for both algorithms, derived from the fits that can be observed in graphs 3b and 4b are

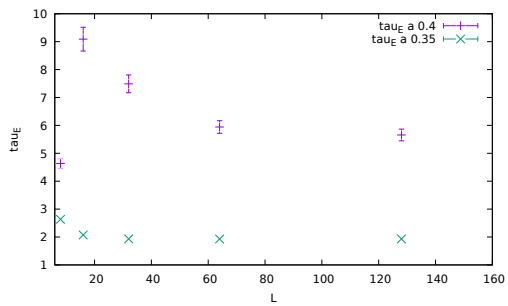
	E		M^2	
	HB	SW	HB	SW
$\tau_{exp,O}$	$2412{,}5 \pm 4{,}5$	$6{,}005 \pm 0{,}033$	$2714{,}5 \pm 1{,}1$	$5{,}748 \pm 0{,}027$

(a) expontential autocorrelation times $\tau_{exp,O}$

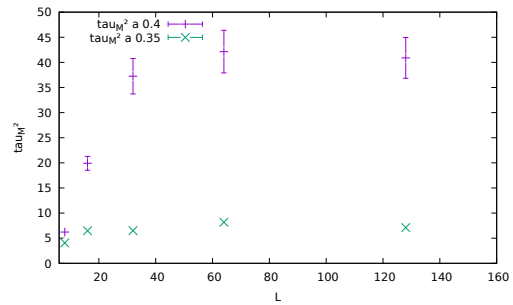
Es It is important to note that the fits were performed for HB in the $2000-4000$ region and in the $5-20$ region for SW, and in both cases a p-value on the order of ~ 0 and a highly deviated χ^2/Dof were obtained, since the correlation function does not return errors in gnuplot and the program is unable to return these values correctly. However, the obtained results correctly express the expected outcome, as it is clearly observed that the exponential autocorrelation time is much greater for HB than for SW.

Finally, we can perform the same study with values of $k < k_c$ and analyze the data; we expect the integrated autocorrelation times to decrease drastically since we are far from the critical point, especially for HB, which is the most affected near the critical point. The simulations are performed

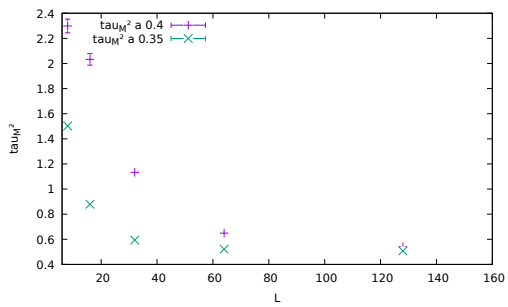
again for $k = 0{,}35$ and $k = 0{,}4$. The results are plotted below in graphs dependent on L. I run SW up to 128 in order to better compare the results.



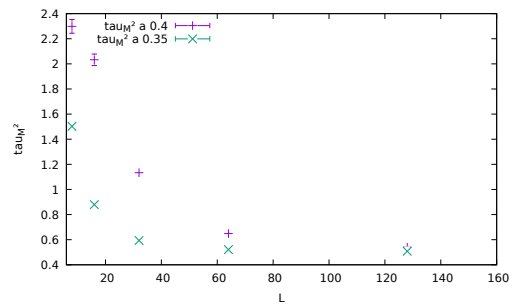
(a) Energy(L) for k<kc (HB)



(b) Square of the magnetization(L) for k<kc (HB)



(c) Energy(L) for k<kc (SW)

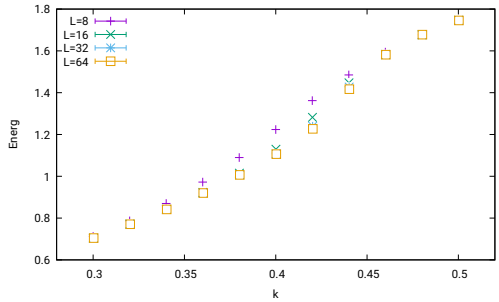


(d) Square of the magnetization(L) for k<kc (SW)

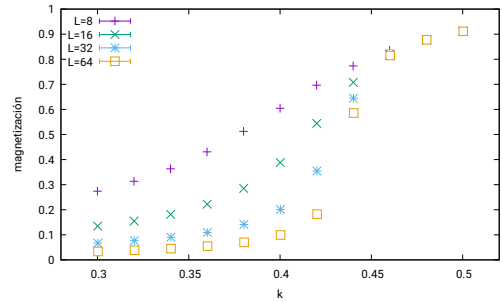
It is indeed verified that as the value of k decreases, the autocorrelation time is lower, by orders of magnitude in the case of HB. The difference between both algorithms remains noteworthy, with SW outperforming by at least one order of magnitude.

3. Measurements of mean values

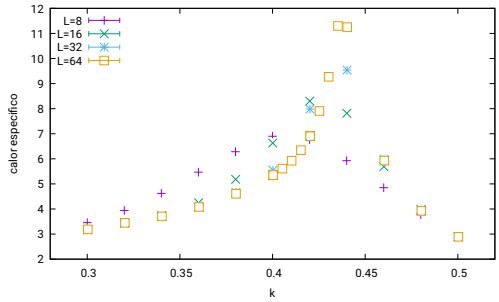
Next, the mean values of various quantities for the Swendsen-Wang algorithm are studied, as it has been verified to be faster in the vicinity of the critical point. We will perform the simulations with an n_{block} and l_{block} that allow us to ensure we reach the thermalization of the system; in this case, for SW, 10×10000 is sufficient. Below, the quantities of interest are plotted from 0.3 to 0.5 in steps of 0.2 from $L = 8$ to $L = 64$. From 0.4 to 0.44, the simulation was performed for $L = 64 - 256$ in steps of 0.05 in order to perform more precise fits later on.



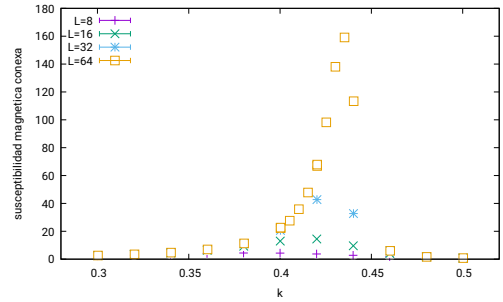
(a) Energy



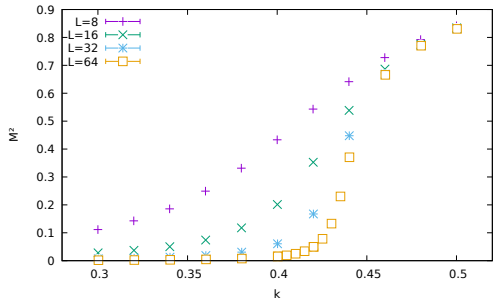
(b) Magnetization



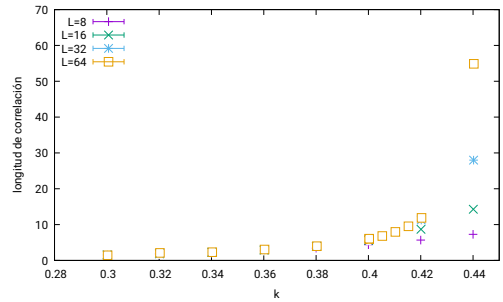
(c) Specific heat



(d) Connected susceptibility



(e) Square of magnetization



(f) Correlation length

Figura 8: Mean values

It can be observed that the energy exhibits a practically linear behavior with a slight curvature as it approaches the critical point (the reason why the specific heat begins to diverge at the critical point), and then recovers its linear behavior after passing the critical point. The magnetization and the squared magnetization exhibit the expected behavior below the critical temperature; however, it can be observed that for smaller lattices, they struggle significantly to approach zero. For the specific heat and the susceptibility, it can be observed that the divergence becomes more pronounced as the lattice size increases, as expected. Furthermore, although it will not be studied in detail in this report, it can be visually appreciated that the position of the maximum shifts to the right as the lattice size increases, until at $L = 128$ the peak is located practically at the true critical temperature. The law governing the apparent critical point is $|T_{c,L} - T_c| \sim L^{-1/\nu} \sim \frac{1}{L}$. Finally, it can be seen in the plot of the correlation length that it diverges increasingly as the lattice size is enlarged and the critical point is approached.As previously mentioned, the exact solution to the Ising model was derived by Onsager. The exact solutions for the mean energy per spin and the absolute value of the magnetization are briefly calculated below to compare them with the results obtained from a simulation using the ising2d program for $L = 2$ at the critical point. Recalling the Hamiltonian for a zero external field 1, for $L = 2$ there is a total of 4 spins and therefore 16 possible configurations. It is important to note that this Hamiltonian possesses inversion symmetry; by analyzing each configuration and determining its contribution to the final value of the Hamiltonian, the partition function can be obtained:

$$Z = \sum_{\langle s_i = \pm 1 \rangle} e^{\beta H} = 2e^{8\beta J} + 2e^{-8\beta J} + 12e^0 = 4cosh(8k) + 12 \tag{5}$$

$$\langle E \rangle = -\frac{\partial lnZ}{\partial \beta} = -\frac{8Jsenh(8k)}{cosh(8k) + 3} \tag{6}$$

Taking into account that what the program will provide us is the energy value per spin, and that we want to calculate it at the critical point, we will have to divide by 4 and substitute $k = k_c$. Obtaining

$$\frac{\langle E \rangle (k_c)}{4} = -1{,}6971 \tag{7}$$

The value provided by the simulation is negative, so the absolute value will be compared. The exact mean value of the magnetization is 0; we study the mean value of the absolute value of the magnetization:

$$\langle |M| \rangle = \langle | \sum_i |s_i| | \rangle = -\frac{\sum_{s_i} |s_i| e^{-\beta E_i}}{Z} \tag{8}$$

and taking into account the 16 possible configurations, dividing by 4 and substituting at the critical point, we obtain

$$\frac{\langle |M| \rangle (k_c)}{4} = \frac{e^{8k} + 2}{2cosh(8k) + 6} = 0{,}88924 \tag{9}$$

Below, I will present in a table the results I have obtained for the simulation, along with the comparison to the theoretical value we just obtained. Se It can be seen that for the energy, a fairly

E		$\lvert M \rvert$	
Simulación con HB	σ_E	Simulación con HB	σ_M
$1{,}6977 \pm 0{,}0027$	0,22	$0{,}8994 \pm 0{,}00084$	11,28

(a) Monte Carlo simulation resultrs for L=2

close value was obtained, within less than one sigma; the value obtained for the magnetization is quite close, but due to the precision of the measurement, the result is more than 10 sigmas away. This can be justified by noting that when performing the simulation at the critical point, the fluctuations of the system are maximal, and the value of the order parameter can vary, leading to this result.

4. Thermodynamic limit

To study the thermodynamic limit, we plot the obtained points along with their theoretical value according to the expression

$$M(k) = (1 - sinh(2k)^{-4})^{1/8} \tag{10}$$

It can be seen 9a how below the critical temperature the simulation overlaps remarkably before reaching the critical temperature, something that does not occur in the disordered phase because the system is not in the thermodynamic limit and, strictly speaking, there is no phase transition. The fluctuations of the order parameter do not exactly cancel each other out. It can be clearly observed how, for larger lattices, these fluctuations do approach zero more rapidly.
On the other hand, we can see that it has been possible to represent in the plot 9b the value of the specific heat in the thermodynamic limit. The results from the book [2] have been employed, in which the expression for the specific heat is derived; it is important to emphasize that gnuplot cannot correctly plot the divergence, and the set samples function has been used to view it approximately. Onsager's solution procedure is described in detail in section 5.5.1, and the result is equation (5.79),

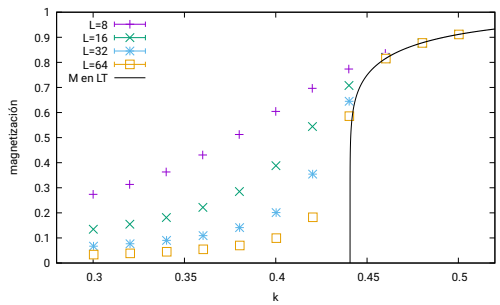
dividing by k^2 to account for the chain rule and the notation, as explained in the help guide for representing this function.

$$c(T) = \frac{4}{\pi}(coth(2k))^2[K_1(\kappa) - E_1(\kappa) - (1 - tanh^2 2k)(\frac{\pi}{2} + (2tanh^2(2k-1)K_1(\kappa))] \tag{11}$$

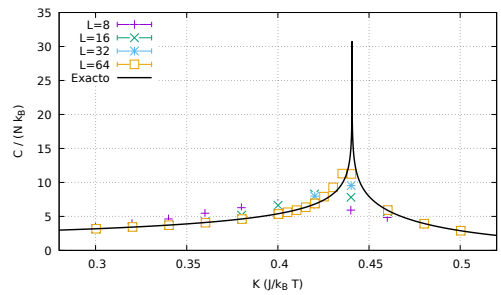
with

$$\kappa = 2\frac{senh(2k)}{cosh^2(2k)} \tag{12}$$

being K_1 y E_1 the elliptic integrals of first and second order respectively.



(a) Thermodynamical limit magnetization



(b) specific heat with thermodynamical limit plot

Se ve It can be clearly seen how the data overlap quite accurately for the largest lattice, and as we approach the critical point, the data move away from the exact value until, at the critical point, they are completely off the plot.
Next, the fit for the inverse of the correlation length is performed. For this section, the inverse of the correlation length has been plotted using data from all lattices, retaining the smallest possible lattice as long as the value at that lattice size was compatible with the next larger one. In this way, a sweep of values for the correlation length has been obtained as we approach the critical point, which can be considered the closest approximation to the thermodynamic limit achievable. For this section, simulation data for $L = 512$ at the point closest to the critical point, 0,435, were required. After obtaining this sweep of values for the inverse of the correlation length, two fits were performed: one ($f(x)$) by fitting to k_c assuming the critical exponent ν is known, and another ($g(x)$) in which the exact value of the critical temperature is assumed to be known and the critical exponent ν is left as a free parameter for the fit. Both fits are represented in a plot and are:

$$f(x) = A \cdot (a - x)^1; \qquad g(x) = B \cdot (k_c - x)^b \tag{13}$$

where a is the parameter that provides the value of the critical temperature and b is the critical exponent associated with the correlation length. Before presenting the results of the fits, it is necessary to emphasize that these fits are designed to hold in the vicinity of the critical point; therefore, when performing the fit, we will restrict ourselves to the values closest to it, even if they are few and we have few degrees of freedom, since moving away from the critical point would render the fit theoretically incorrect. The following is obtained:
Por Finally, I will study how the position of the maximum shifts as a function of L for the specific heat and the connected magnetic susceptibility. Firstly, the connected magnetic susceptibility in the thermodynamic limit should scale ascomo [1]

$$\chi \sim A{\cdot}L^B; \qquad B_{teórico} = \frac{\gamma}{\nu} = 7/4, \tag{14}$$

let us see what happens if we perform a fit and whether the value of 7/4 for the exponent is actually obtained. On the other hand, we know that the specific heat must scale logarithmically according to

$$c \sim a \cdot logL + b \tag{15}$$

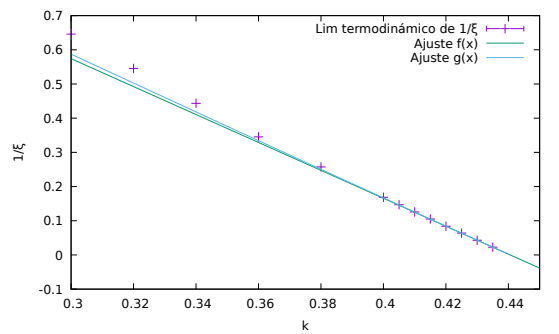
which implies that each time the lattice size is increased, the value of the specific heat would increase by a constant amount. Note that we are using the peak values as the maximum values

k_c	$k_{c,err}$	σ_{k_c}	χ^2/Dof	p-value
0,440557	0,000050	2,57	1,07	0,34

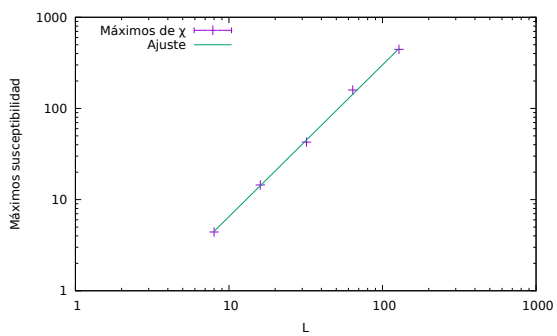
(a) fit to k_c, f(x)

ν	ν_{err}	σ_ν	χ^2/Dof	p-value
1,0138	0,0053	2,57	0,96	0,38

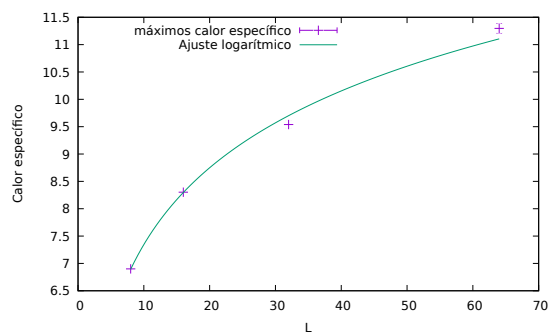
(b) fit to critical exponent ν, with g(x)



(c) fit to the inverse of correlation length



(a) fit to the max points of susceptibility



(b) fit to the max points of heat capacity

obtained from the simulations represented in the plots. 8c and 8d Although graphically the fits may appear reasonably correct, p-values of 0 and 0,0027 were obtained, respectively, indicating that the expected relations are not being strictly satisfied. The primary explanation for this discrepancy is the systematic error incurred by assuming that the maximum values obtained in our simulation (sampled only at certain discrete values of k) correspond to the true maximum of each function. It would have been more accurate to extrapolate the mean values to locate the true maximum and perform the fit using those peaks. To illustrate the error in these fits, in addition to the p-values, I present below the value obtained for the exponent $B = 1{,}665 \pm 0{,}023$, a result significantly far from the theoretical value of 7/4 due to the aforementioned reasons.

5. Finite Size Scaling

Finite Size Scaling is the study of phase transitions and critical phenomena observed in the real world through systems that are finite and possess a characteristic dimension L. We perform this study because, as seen in previous sections, intensive quantities depend on the lattice size. Consequently, we propose an ansatz in which the correlation length is limited by the physical dimensions of the system, such that the finite-size behavior is governed by the ratio $\frac{L}{\xi_\infty}$. [1],where the denominator is the correlation length for an infinite-sized system. Thus, the observables are described through the

expression

$$\langle O \rangle_L = L^{x_0/\nu} f_O(\frac{L}{\xi_\infty}) \tag{16}$$

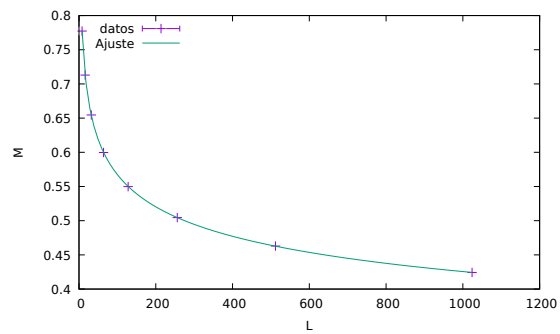
where f_O is an analytical function dependent on the boundary conditions and is close to zero. This formalism allows taking into account the effects of being in a finite-sized system and calculating the critical exponents associated with each of the observables.
We begin by studying the critical exponent β, which is related to the magnitude of the magnetization through the expression:

$$M \sim (T_c - T)^\beta \Rightarrow M \sim A \cdot L^{-\beta} \tag{17}$$

Without taking into account L=8 we obtain:

β	β_{err}	σ_β	χ^2/Dof	p-value
0,12484	0,00015	1,05	0,88	0,49

(a) Critical exponent β



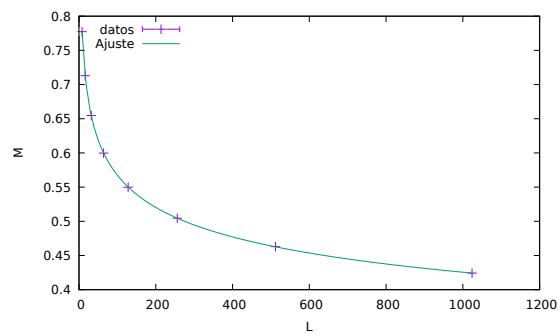
(b) Magnetization fit

Un quite a close result; however, to illustrate the power of FSS, if we take into account higher-order FSS corrections to attempt to improve the results, the fit is

$$M \sim A \cdot L^{-\beta} + B \cdot L^{-c} \tag{18}$$

β	β_{err}	σ_β	χ^2/Dof	p-value
0,12496	0,00070	0,051	1,06	0,37

(a) Critical exponent β with first order corrections



(b) Magnetization fit (first order correctiones)

Obtaining a result that is practically the exact value, as indicated by being within $< 0{,}1$ sigmas.
To study the critical exponent η, we perform the power-law fit on the squared magnetization, which is given by the expression

$$M^2 \sim A \cdot L^{-\eta} + B \cdot L^{-2} \tag{19}$$

and with every length on the fit we get:
To calculate the critical exponent ν, we use the derivative of the correlation length, as it will provide the necessary information to perform a fit if we study it at the critical temperature. Since as

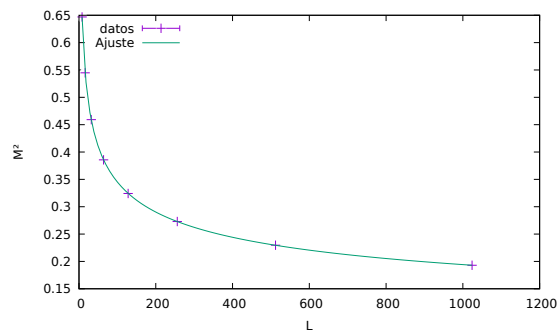
$$\xi = L^{\nu/\nu} f(L^{1/\nu}(\kappa - \kappa_c)) \tag{20}$$

with derivative:

$$\frac{d\xi}{d\kappa_c} = L L^{1/\nu} f'(L^{1/\nu}(\kappa - \kappa_c)) = L^{1+1/\nu} f'(0) \tag{21}$$

η	η_{err}	σ_η	χ^2/Dof	p-value
0,24977	0,00029	0,78	0,77	0,57

(a) Critical exponent η



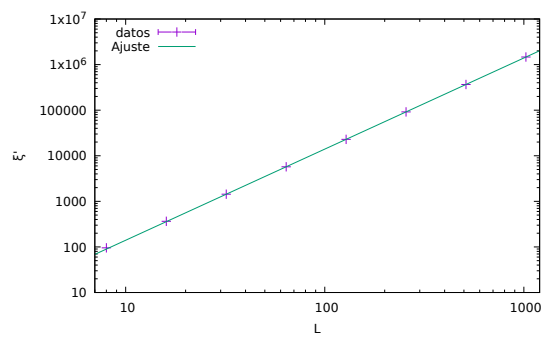
(b) Square of the magnetization fit

and since the derivative of the scaling function evaluated at the critical point is constant, we find that we can fit to the power law:

$$\xi' \sim A \cdot L^b; \qquad b = 1 + \frac{1}{\nu} \tag{22}$$

ν	ν_{err}	σ_ν	χ^2/Dof	p-value
1,0007	0,0010	0,685	0,14	0,97

(a) Critical exponent ν



(b) Correlation length derivative fit

To perform this fit, I had to discard the data from the lattices of size 8 and 16, since the correlation length, taking into account the first-order correction, is:

$$\xi = L^b(1 + \frac{B}{L^{1,75}} + \cdots) \tag{23}$$

in which the term B is relatively large; therefore, if the lattice is very small, the correction is not negligible. For this reason, to calculate the exponent ν, the smallest lattices had to be discarded. It can be verified that the correlation length scales linearly (see the obtained fit parameters) if we perform the fit and remove lattices 8 and 16 for the same reason explained for obtaining ν, which is justified by23,

$$\xi \sim aL \tag{24}$$

The critical exponent γ is given by the power law followed by the connected susceptibility

$$\chi_c \sim A \cdot L^\gamma \tag{25}$$

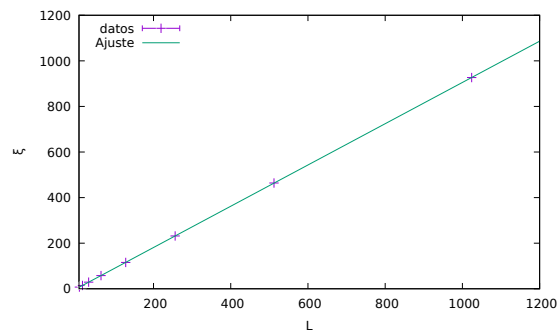
And to perform this fit, the $L = 8$ lattice had to be discarded in order to obtain an acceptable p-value (otherwise it is of the order of $\sim 10^{-5}$), such that:
Finally, the critical exponent α remains to be studied. This critical exponent is related to the specific heat through the expression:

$$c_e = A \cdot L^{-\alpha} \tag{26}$$

However, if we perform this fit, even if we remove the values from some of the lattices, we find that the fit does not follow a power law, as a p-value $= 0$ is obtained for any fit configuration, along with

χ^2/Dof	p-value
1,64	0,144

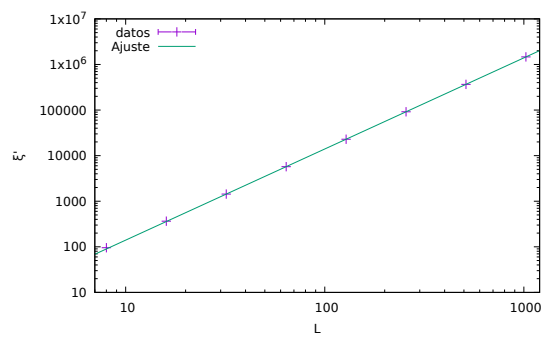
(a) fit values



(b) correlation length fit

γ	γ_{err}	σ_γ	χ^2/Dof	p-value
1,75176	0,00077	−2,29	1,29	0,27

(a) Critical exponent γ



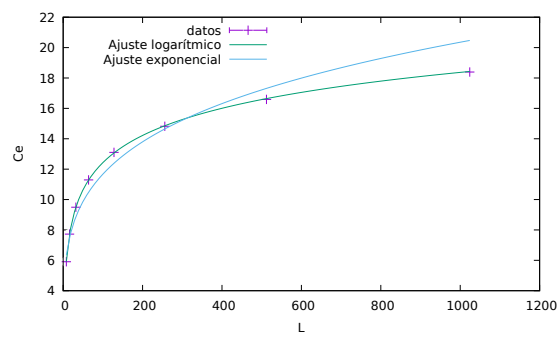
(b) Connected susceptibility fit

a $\chi^2/\text{Dof} = 1241{,}51$ when considering all lattices. From these results, it can be concluded that this observable does not follow a power law and therefore is 0. In fact, if we attempt to perform a fit using a logarithmic law, we obtain:

$$c_e = A + B \cdot logL \tag{27}$$

χ^2/Dof	p-value
1,41	0,20

(a) fit values



(b) specific heat fits

De the fit parameters allow us to conclude that the law followed by the specific heat is logarithmic and not exponential. Thus, the critical exponent is zero, $\alpha = 0$.
To find the critical exponents δ and α numerically, we must use the relations between the critical exponents. These relations are the Widom relation

$$\gamma = \beta(\delta - 1) \tag{28}$$

and the Griffiths one:

$$\alpha + \beta(1 + \delta) = 2 \tag{29}$$

, the error can be obtained using the usual propagation of errors (assuming that the variables are independent and have zero covariance). Thus, we obtain:

δ	δ_{err}	σ_δ
15,019	0,079	0,24

(a) δ a través de la relación de Widom

α	α_{err}	σ_α
−0,002	0,015	0,11

(b) α a través de la relación de Griffiths

Values that, as can be observed, are within less than 1 sigma. Finally, we can study the spatial dimension of the problem to conclude checking that the results are coherent; the relation is

$$d = \frac{2-\alpha}{\nu}, \tag{30}$$

obtaining a result of $d = 2{,}001 \pm 0{,}015$. Therefore, it can be concluded that the study of the critical exponents is satisfactory, as the results obtained match the expected values dictated by the exact solution of the Ising model. All critical exponents are within less than 1 sigma, except for γ, partly due to its very small associated error.

6. Ising Model applied to financial markets

To conclude this report, I will examine whether an analogy can be drawn between the previous study and financial markets. The objective is to verify if the critical exponents obtained from 2017 and 2018 data coincide with those of a 2D Ising model and, consequently, if the market can be studied as a block of interacting spins. I will use data from companies listed on the S&P 1500 during those two years.

To carry out this process, I will consider an üp"spin to mean that the stock price increased compared to the previous day, and a "down"spin if the closing price was lower than the day before. I will also assume a constant interaction coupling between all companies. Two studies of the market are performed: the first assumes the market is a system constantly at the critical temperature, and the second uses the VIX index—which measures market volatility (in this case, for the S&P 500)—to select only those days where volatility falls within a threshold between different percentiles (70-95 was tested). To draw an analogy with financial markets, I will interpret temperature as a measure of noise and/or volatility. By this, I mean that in reality, being affected by news and official corporate communications, the financial market tends to move in blocks according to sector. If state subsidies for oil are announced, all oil companies rise, etc. Based on this, the following ansatz is proposed: if we arrange the companies by sector blocks (banks, insurance, pharmaceuticals, etc.) in a square two-dimensional grid and assume we are at the critical temperature, the correlation length should be infinite (or larger than the system size to apply FSS), and the system should behave like the set of spins observed in previous sections. As expected, it will be shown that this is not the case, but conclusions can be drawn on how to model the financial market and create more precise models that might explain the universality of this system, if possible.

The data was extracted from historical listings of companies included in the S&P 1500 index (which comprises the S&P 500, S&P 400 MidCap, and S&P 600 SmallCap indices) between the years 2017-2018. [3, 4, 5]and the closing price data were extracted using the Yahoo Finance Python API. [6]. Following this, fits were performed in the same manner as in previous sections, but using Python. The code follows the logic of the ising2d.c program (for the Swendsen-Wang cluster Monte Carlo method) to obtain parameters such as magnetization, specific heat, and the variables necessary to perform the fits for the critical exponents. Data from 1,500 companies were required to apply the finite size scaling fits across at least three lattice lengths, of sizes $L = 8, 16, 32$. It would have been more useful for obtaining the critical exponents to have a range that allowed for larger lattices, but obtaining such data via APIs requires paid tools that are not available. The Monte Carlo method code applied for the simulation and the analytical derivation of the variables will be available in the GitHub repository ; for the analytical derivation, we use[1] and follows from equations 6 and 8 . The critical exponents were obtained in this case using linear fits by taking logarithms of each of the magnitudes, as shown to be possible in equation 2-12 of [1]

$$\langle O \rangle_L = L^{x_O/\nu} f_O(L/\xi_\infty) \tag{31}$$

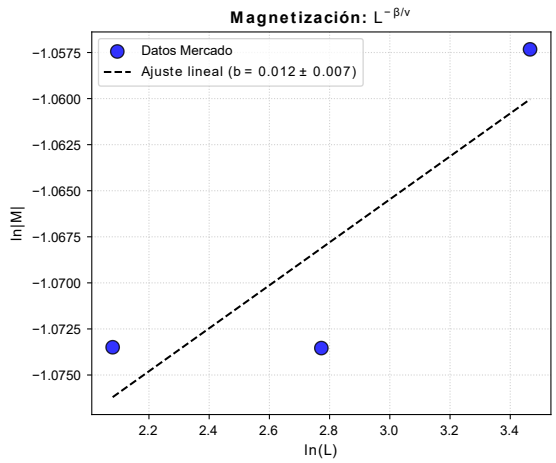
Obtaining the ratio x_O/ν as the fit parameter. For the exponent ν, the Rushbrooke relation obtained directly from 28 and 29

$$\alpha + 2\beta + \gamma = 2 \tag{32}$$

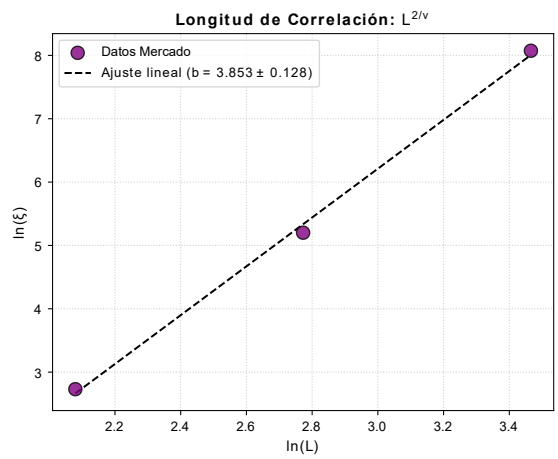
and using 31 you obtain

$$\nu = \frac{2}{(\frac{\gamma}{\nu})_{ajuste} + 2(\frac{\beta}{\nu})_{ajuste} + (\frac{\alpha}{\nu})_{ajuste}} \tag{33}$$

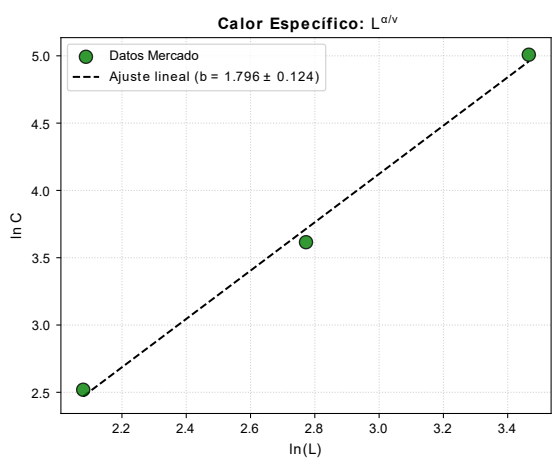
So u can derive 14



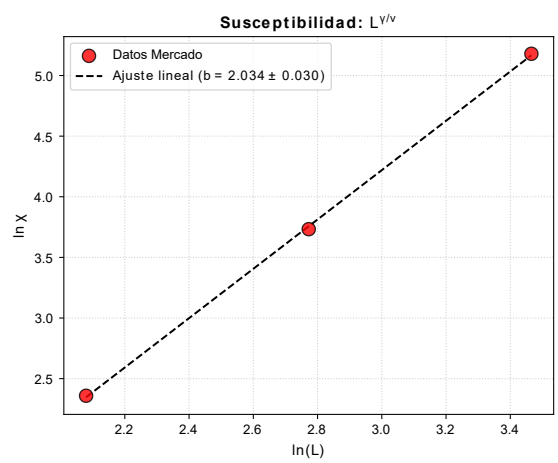
(a) Magnetization fit



(b) Correlation length



(c) Specific heat fit



(d) Susceptibility fit

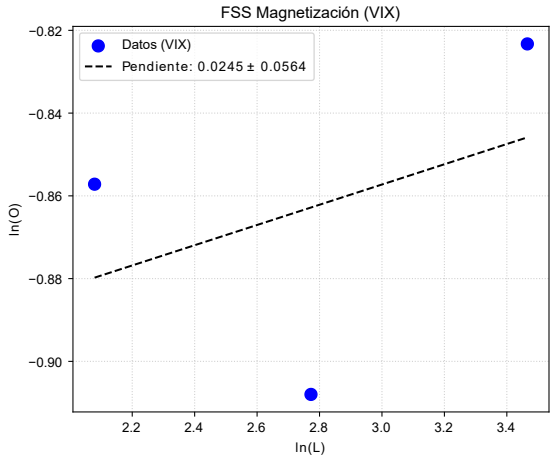
Cuadro 14: Fits

After data processing and the usual error propagation, the results were obtained. Following an initial acquisition, they were compared with theoretical values, not only those of Onsager but also those derived from mean-field theory.

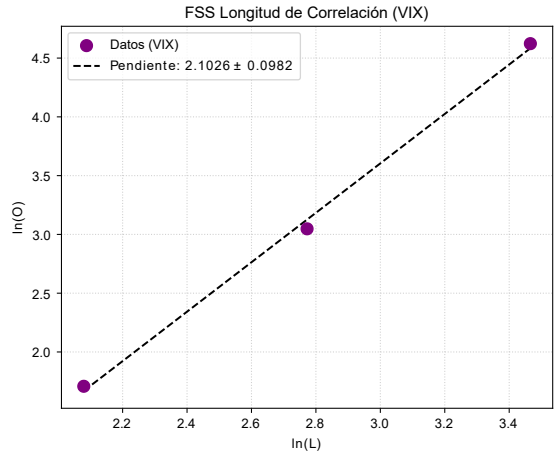
Exponente	**Valor Ajustado**	**Onsager**	**Campo medio**
ν (Long. Corr)	$0{,}519 \pm 0{,}017$	1,00 (27,9σ)	0,50 (1,1σ)
α (Calor)	$0{,}932 \pm 0{,}071$	0,00 (13,1σ)	0,00 (13,1σ)
β (Mag)	$0{,}006 \pm 0{,}004$	0,12 (33,8σ)	0,50 (140,3σ)
γ (Susc)	$1{,}056 \pm 0{,}038$	1,75 (18,1σ)	1,00 (1,5σ)

Cuadro 15: Comparación de valores ajustados frente a los modelos de Onsager (2D) y Campo Medio.

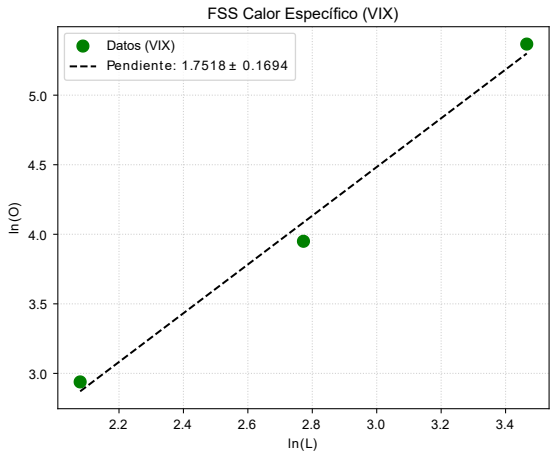
The following are the results obtained after filtering the data using the volatility index (VIX), retaining only the days between the 70th and 95th percentiles of the index. This is performed to determine whether, by considering days with high susceptibility, the universality of the two-dimensional Onsager solution holds, as we are isolating only those days at the 'critical temperature'. The data up to the 100th percentile are not considered to avoid days where volatility was extremely high due to major news events. I assume that on those days, the market does not exhibit normal behavior as it is influenced by what could be interpreted in the magnetic model as an external magnetic field. The results are obtained using the same method as the previous ones without the VIX index filtering. 16



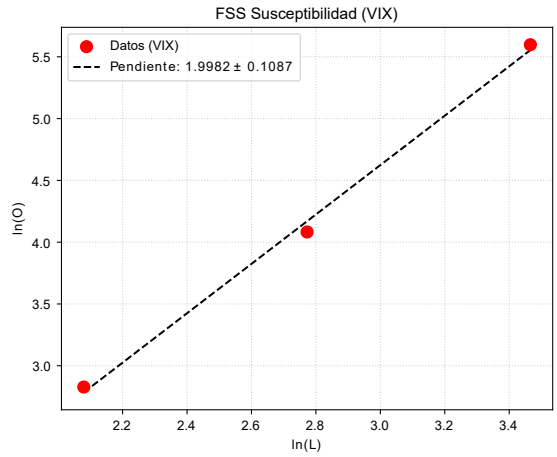
(a) Magnetization fit VIX filtered



(b) Correlation length VIX filtered



(c) Specific heat fit VIX filtered



(d) Susceptibility fit VIX filtered

Cuadro 16: Filtered fits

Exponente	**Valor Ajustado(VIX)**	**Onsager**	**Campo medio**
ν	$0{,}526 \pm 0{,}032$	1,00 (14,8σ)	0,50 (0,8σ)
α	$0{,}922 \pm 0{,}105$	0,00 (8,8σ)	0,00 (8,8σ)
β	$0{,}013 \pm 0{,}030$	0,12 (3,8σ)	0,50 (16,4σ)
γ	$1{,}052 \pm 0{,}086$	1,75 (8,1σ)	1,00 (0,6σ)

Cuadro 17: Comparación de exponentes críticos filtrados (VIX) frente a soluciones de Onsager (2D) y Campo Medio.

The tables show the standard deviations (σ) by which both studies differ from the theoretical values of Onsager and Mean Field theory. It can be observed in both tables that the values obtained for the critical exponents deviate significantly from the theoretical values of the Onsager solution. This concludes the idea that the financial market, modeled through nearest-neighbor interactions at a presumed critical temperature, behaves like a two-dimensional block of spins. As previously mentioned, this was expected, as many factors are not being taken into account that this Ising Hamiltonian—without an external field—cannot capture. Specifically, the real interaction is not limited to nearest neighbors but includes positive correlations for some assets and negative ones for others (competing companies), and we have not added an external field to represent news or actual market trends.

On the other hand, after performing the fits and obtaining the exponents, it is striking how the values obtained for ν and γ are very similar to the mean-field exponents. However, this could have been anticipated, as it suggests that closing price correlations behave as if each company were connected to a global upward or downward movement, which can be translated into an effective mean field. Similarly to the correlation, the susceptibility exhibits behavior akin to mean field, scaling linearly as we approach the system's critical point.

Nevertheless, we find that α and β have values that do not correspond to either theory. These exponents indicate that the magnetization has a discontinuity near the critical point and that the specific heat possesses a divergence. These behaviors are inconsistent with those observed for a ferromagnetic phase transition and indicate that we are far from explaining financial market behavior through a two-dimensional Ising Hamiltonian without an external field.

Referencias

[1] Daniel J. Amit and Victor Martin-Mayor. *Field Theory, the Renormalization Group, and Critical Phenomena: Graphs to Computers*. World Scientific Publishing Company, Singapore, 3rd edition, 2005.

[2] H. Gould and J. Tobochnik. *Statistical and Thermal Physics: With Computer Applications, Second Edition*. Princeton University Press, 2021.

[3] Wikipedia contributors. List of s&p 400 companies — Wikipedia, the free encyclopedia, 2026. [Online; Data retrieved: 25-March-2026].

[4] Wikipedia contributors. List of s&p 500 companies — Wikipedia, the free encyclopedia, 2026. [Online; Data retrieved: 25-March-2026].

[5] Wikipedia contributors. List of s&p 600 companies — Wikipedia, the free encyclopedia, 2026. [Online; Data retrieved: 25-March-2026].

[6] Yahoo Finance. Historical market data api, 2026. [Online; Data retrieved via yfinance Python library: 25-March-2026].